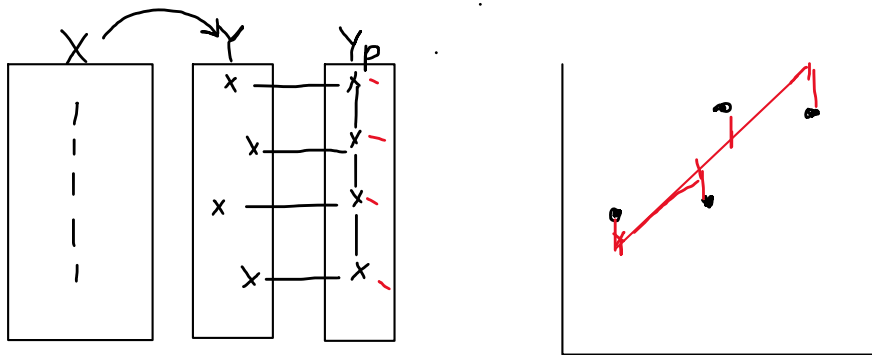
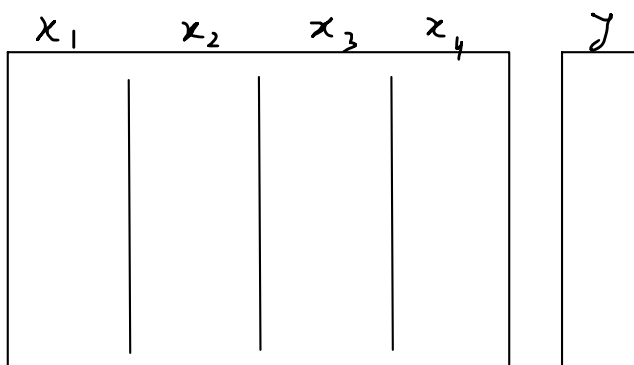




Loss = mean_squared_error

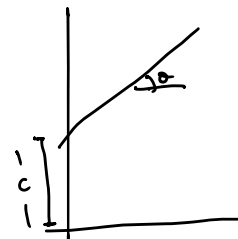


$$Y_p = w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + \text{bias}$$

→ $Y_p = w_1x_1 + \text{bias}$

$$y = mx + c$$

$$\begin{matrix} \text{[]} & = & \text{[]} \times \text{[]} & + & \text{[]} \\ & & \text{[]} & & \end{matrix}$$



cleaning

← pandas

feature scaling

← sklearn.preprocessing

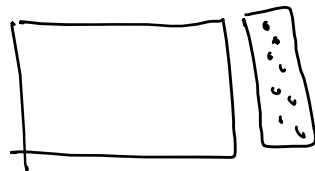
model

linear_model → LinearRe
- svm → SVM

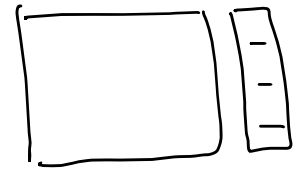
Modeling
Estimate

← sklearn.
 ↗ linear_model → LinearRe
 ↘ svm → SVMs
 ↘ neighbours → KNN
 ↘ tree → DecisionTree

Classification



Regression



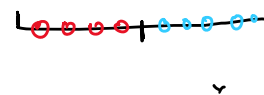
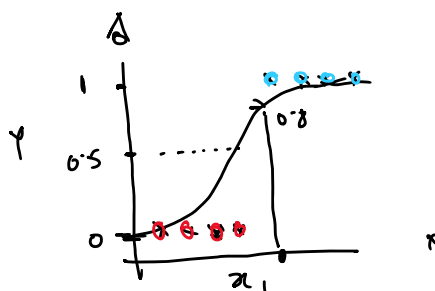
Classification

Logistic Regression

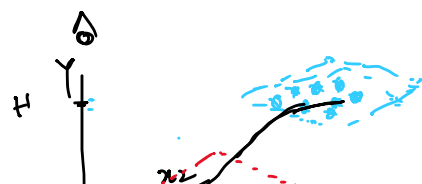
$$y_p = \sigma \left(\underbrace{w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4}_{z} + b \right)$$

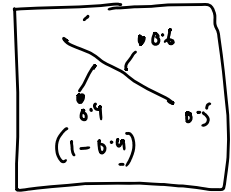
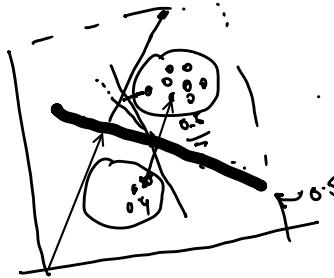
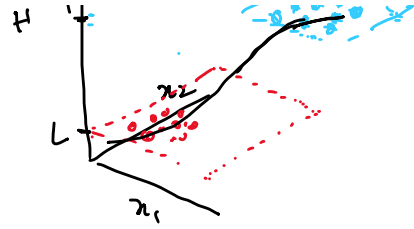
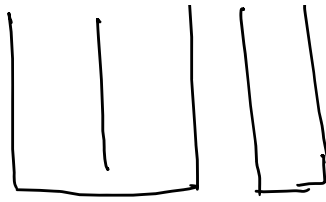
$$\frac{e^x}{1+e^x} = \frac{e^{w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + b}}{1 + e^{w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + b}} = \frac{e^z}{1+e^z}$$

x_1	x_2	x_3	x_4	Y	z	y_{pred} σ	y_{real}
				1	-2.2	0.4	0
				1	+1.8	0.6	1
				1		0.7	1
				1		0.9	0



x_1	x_2	Y





$$\log_e(e^x) \rightarrow x$$

$$\left. \begin{aligned} x^2 &= N \\ 2 &= \log_2 N \end{aligned} \right\}$$

$$\left. \begin{aligned} e^k &= N \\ k &= \log_e N \\ \log_e e^k &\rightarrow k \end{aligned} \right\}$$

$$y = e^{y_1}$$

$$(1-y) = e^{(1-y)_1}$$

$$y \times \log(y)$$

$$(1-y) \times \log(1-y)$$

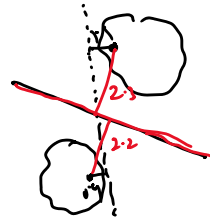
$$\downarrow$$

$$(1-0)$$

$$y \in \{0, 1\} \text{ loss function} = - \sum \left(\frac{y \times \log \sigma}{\downarrow} + \frac{(1-y) \log (1-\sigma)}{\downarrow} \right)$$

$$\log(0.01) \rightarrow -2$$

y	σ	
0	0.3	0.7
1	0.8	0.2
0		
1		
0		



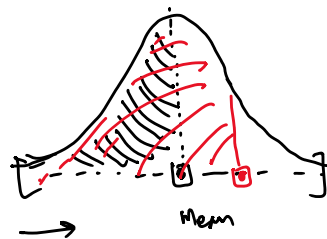
$$\therefore \log(0.02) = -ve = +ve$$

$$\text{Neg Log Likelihood Loss}$$

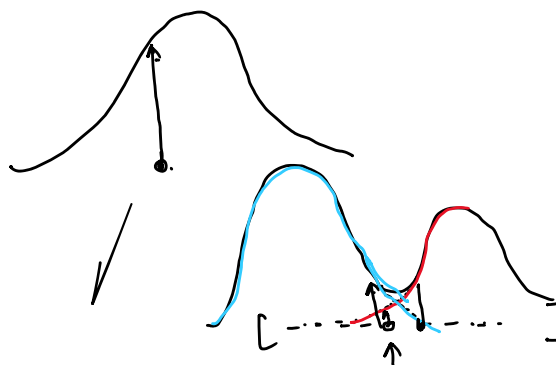
$$- \log \frac{y(\sigma) + (1-y)(1-\sigma)}{y(\sigma) + (1-y)(1-\sigma)}$$

Gradient Descent

Loss function



50% → 0.5
80% → 0.8



Graph

y	$p(i)$	Likelihood (L)	L
0	0.3	$1 - 0.3$	0.7
1	0.7	0.7	0.7
0	0.2	$1 - 0.2$	0.8
1	0.9	0.9	0.9
0	0.1	$1 - 0.1$	0.9
1	0.6	0.6	0.6
			Σ

$$-\frac{1}{N} \sum_{i=1}^N y_i \cdot \log(p(y_i)) + (1 - y_i) \cdot \log(1 - p(y_i))$$

Binary Cross-Entropy / Log Loss

Y	σ	Likelihood	Log Likelihood
Actual	Predicted Probabilities	Corrected Probabilities	Log
1	0.94	0.94	-0.02687
1	0.9	0.9	-0.04576
1	0.78	0.78	-0.10791
0	0.56	$1 - 0.56$	-0.35655
0	0.51	$1 - 0.51$	-0.3098
1	0.47	0.47	-0.3279
1	0.32	0.32	-0.49485
0	0.1	$1 - 0.1$	-0.04576